COMPLETE GLOBAL EMBEDDEDNESS OBSTRUCTION FOR THE RECIPROCAL-REFLECTION KATO–MUROYA GENUS-ONE FOUR-CATENOIDAL-END LOCUS

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove a complete global embeddedness obstruction for the normalized nonzero-spin Kato–Muroya genus-one four-catenoidal-end Weierstrass locus admitting reciprocal reflection $I(z) = 1/\bar{z}$ and a common unoriented limiting-normal axis. Covering every rectangular elliptic modulus $r > 1$, both signed half-periods, and all nine puncture distributions $N_1 + N_2 + 2N_3 = 4$ (both the fixed-puncture strata $N_1 + N_2 > 0$ and the free-pair stratum $N_1 = N_2 = 0,\ N_3 = 2$), we prove that no regular degree-four datum in this locus yields a globally embedded minimal surface in $\mathbb{R}^3$. Furthermore, using similarity-invariant grouped end-growth invariants and horizontal handle flux, we show that the excluded four-torsion period circle strictly extends the classical symmetric Wohlgemuth CCCC family.

1. Main Mathematical Results

Theorem 1.1 (Complete reciprocal-reflection genus-one four-end obstruction)**.** *Across all elliptic moduli $r > 1$, both signed half-periods, and all nine puncture distributions $N_1 + N_2 + 2N_3 = 4$, the normalized nonzero-spin Kato–Muroya genus-one four-catenoidal-end locus with reciprocal reflection $I(z) = 1/\bar{z}$ and a common unoriented limiting-normal axis contains no globally embedded minimal surface in $\mathbb{R}^3$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 6800022 in the domain of differential geometry and minimal surfaces adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Complete nonexistence of globally embedded minimal surfaces across all nine puncture distributions N_1+N_2+2N_3=4, all moduli r > 1, and both signed half-periods in the normalized nonzero-spin reciprocal-reflection Kato-Muroya genus-one four-end locus.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted existence of a complete properly embedded genus-one four-ended minimal surface without symmetry assumptions

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 6800022 (differential geometry and minimal surfaces), archived in the accompanying repository package `results/6800022/`.

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